Innovation Corps for Learning (I-Corps-L):
Evidence-based Entrepreneurship™ to Improve (STEM) Education

Principal Investigator- Karl Smith, Ph.D.

Designed by Kelly Auxier

"Enacting educational transformation by taking an entrepreneurial approach"

Project Description

The I-Corps for Learning (I-Corps-L) is a pilot initiative to propagate and scale educational innovations. The I-Corps-L program is based on the National Science Foundation (NSF) I-Corps program, which guides teams using established strategies for business start-ups, such as the Business Model Canvas and the Lean Launchpad, to build entrepreneurial skills that will encourage mainstream application of their emerging technologies.

Project Goals

- Give the I-Corps-L team an experiential learning opportunity to help determine the readiness of their innovation for sustainable scalability. Sustainable scalability involves a self-supported entity that is sustainable and systematically promotes the adoption of the educational innovation and enables and facilitates its use.
- Enable the team to develop a clear go/no go decision regarding sustainable scalability of the innovation.
- Develop a transition plan and actionable tasks to move the innovation forward to sustainable scalability, if the team decides to do so.

The Overarching goal of I-Corps-L is to foster an entrepreneurial mindset within the education community to impact the way innovations are designed and implemented.

Project Organization

Logistics

Norman Fortenberry
American Society of Engineering Education

Rocio Chavela
Manager of Faculty Development (ASEE)

Leah Smith
Project Coordinator (STEM)

Developers

Steve Blank
Blank's Business Canvas Model

Jerome Engel
I-Corps National Faculty Director

Don Millard
National Science Foundation

Lead Instructors

Karl Smith
University of Minnesota

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Co-Instructors

Shawn Jordan
Arizona State University

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Arizona State University

Russell Korte
Colorado State University

Robert MacNeal
Working Company

Brandy Nagel- TA
Consultant

I-Corps-L Pilot Teams

Mobile Hands-On STEM Pedagogy	Innovative approaches to learning enabled by Mobile Hands-On STEM Pedagogy.
Concept Warehouse	Platform that improves teaching effectiveness by speeding the propagation of evidence-based instructional practices among STEM faculty.
Virtual PLC	Web-based hands-on tool to understand automated systems and controls fundamentals, practice real-life applications and solving automation problems using Programmable Logic Controllers (PLCs).
Math Jam	Week-long school-sponsored boot camp for the Math Placement Test and/or upcoming math courses.
ELeVATE	Holistic transition program that supports veterans interested in engineering and technology careers and educates faculty and staff on how to set veterans up for success in college and beyond.
Engineering Ambassador Network	Program to launch undergraduates into leaders.
Hands-on Programming Experiences	Hands-on coding- microcontroller programming target as base programming experience for all engineering students.
Women in Computing	Institutionalizing successful project to benefit women and Universities nationwide.
Increase the Impact	Providing NSF and PIs ways to design educational development projects that are more likely to result in broader impacts.

Team Members
- Principal Investigator (PI)
- Entrepreneurial/ Administrative Lead (EL)
- Mentor (M)

Evaluation Team

Gary Lichenstein · Cathleen Simons · Sheri Sheppard

QED- Quality Evaluation Designs

Development Program & Resources

7 Week Development Course

3 Day Workshop

-Teams are introduced to the Lean LaunchPad approach

-Teams learn the business model development and customer development process

-Teams meet with customers and ' present what they learned to the class.

Blank's Lean Launch Pad 5 Week Online Workshop

During this five-week period, teams are required to get out of the building and test their business model assumptions, meeting with about 15 customer contacts each week (with the objective of conducting at least 100 total contacts).

Each online class has two parts:
1.5 hours: Team presentations.
1 hour: Class discussion of the weekly lecture.

2 Day Workshop

-Teams present the lessons learned in their exploration of sustainable scalability.

Business Model Canvas (BMC)

Mobile Hands-On STEM Pedagogy

This final BCM lead the team to develop a hands- on learning ecosystem so students can have an experience like that of an engineer when using their products

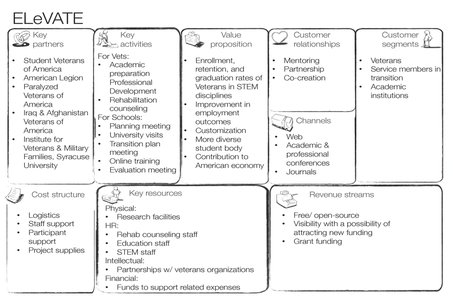
Key Partners	Key Activities	Value Propositions	Customer Relationships	Customer Segments
Hardware/Software Vendors: Digilent, NI, Mathworks, ADI ... Working engineers MOHS Partners & Early Adopters: SIPI, UW, BU, UNM ... HBCU Engineering US & International companies, orgs, etc promoting STEM educ & a diversity . SMART LIGHTING & Other ERCs	Documentation of learning. Hardware agnostic human interface for mobile learning platforms. PCs to tablets/phones ... **Key Resources** Software/App developers Website/Library for sharing content & tools, talk to users ... Eng. Education Researchers	★ **Liberates circuits and electronics learning from cost, time and space constraints.** ★ **Students have expanded opportunities to obtain help with difficult concepts, develop deeper understanding and apply what they have learned in outside activities.** ★ **Instructors have expanded opportunities to add hands-on activities to their courses and to provide help to students who struggle.**	All customers expect ease of use, low cost, access to relevant learning materials and are willing to help one another and add to the overall knowledge base. **Channels** Website & Social Community Workshops, Webinars, Conferences, Journal Articles Magazine Articles	Passionate Instructors in ECE Circuits & Electronics courses who will do what is necessary to help their students. ★ **Instructors in ECE C&E courses who are set in their ways.** Students in ECE C&E courses who have few difficulties with present courses. ★ **Students in ECE C&E courses who need extra help in present courses.** Passionate Instructors in other STEM electronics courses who will do what is necessary for their
Cost Structure			**Revenue Streams**	
Mobile hands-on tools must cost less than textbook. Educational units must provide some limited infrastructure. Component costs subsidized by manufacturers. Content & tools developed by working engineers.			Support from industry interested in improving workforce. In kind support from vendors and maintenance of additional online support materials. Student fees Foundation Support	

Highlighted Team Discovery

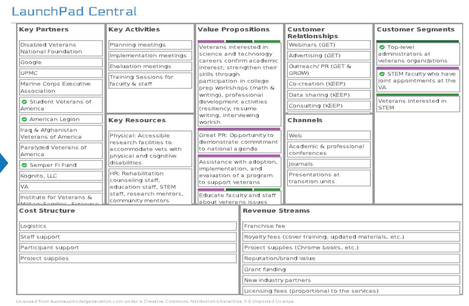
ELeVATE

Holistic transition program that supports veterans interested in engineering and technology careers and educates faculty and staff on how to set veterans up for success in college and beyond.

Data Collected
102 interviews



Initial Business Model Canvas



Final Business Model Canvas

Key Insights
Presented in their lessons learned video

The business model canvas and Lean Launch Pad Central helped determine the key customers for ELeVATE.

100 surveys were necessary to understand market needs.

Value propositions mean different things to each of the customer segments.

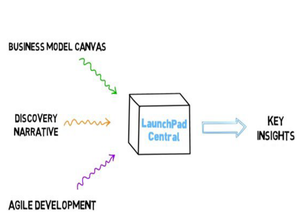
Some of the hypotheses about implement and scaling ELeVATE were valid.

More local work is necessary.

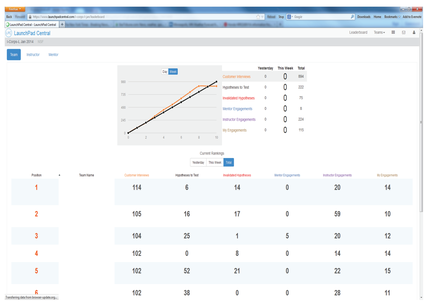
The process provided a mental shift that allowed the team to properly package and provide support to each institution adopting the program.

Lean Launch Pad Central

Lean Launch Pad Central is an online tool used by both the teams and instructors to track progress through the discovery process and indicate key insights based on customer interviews and reiterations of their business model canvas.



Lean Launch Pad Central provides an "X-ray vision into startups using the platform for Evidence based Entrepreneurship" by "changing the way we invest in startups, teach entrepreneurship, run startups, and think about commercializing innovation."



Instructor view of team progress through 5 week program

I-Corps for Learning

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